

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

- (1) Let m be the Lebesgue measure on $[0, 1]$ and $I_A(x)$ denote the indicator function of A .

(a) Consider for $n \geq 1$, $f_n(x) = nI_{[0, \frac{1}{n}]}(x)$ and $f(x) = 0$ for $x \in [0, 1]$. Show:

(i) $f_n \rightarrow f$ a.e.(m).

(ii) f_n is **not** uniformly integrable (UI).

(iii) Check $f_n \not\rightarrow f$ in L^1 .

(b) In general, under what conditions on the functions $\{f_n\}$ and f , would

" $f_n \rightarrow f$ a.e. (m)" imply " $f_n \rightarrow f$ in L^p ($0 < p < \infty$)"?

(c) Show that $f_n \rightarrow f$ in L^p implies that $f_n \rightarrow f$ in L^q , for any $0 < q < p < \infty$.

((3×6)+6+8=32 points)

- (2) Suppose X is a random variable defined on some probability space $(\Omega, \mathcal{F}, P)$.

(a) If $M(t) = E(e^{tX})$, $t \in \mathbb{R}$ is the moment generating function of X , then show

$$P(X > a) \leq \inf_{t>0} e^{-ta} M(t), \quad a > 0.$$

(b) If $X \geq 0$ is random variable with $Var(X) < \infty$. Prove:

$$P(X > 0) \geq \frac{[E(X)]^2}{E(X^2)}.$$

[Hint: Use Cauchy-Schwarz Inequality]

(8+10=18 points)

- (3) (a) State precisely the Dominated convergence Theorem (DCT) (Not the EDCT!).

(b) Prove that the conclusion of DCT remains true even if " $f_n \rightarrow f$ a.e. (μ)" is replaced by " $f_n \rightarrow f$ in measure" in the statement.

(6 + 9 = 15 points)

- (4) Let $(\Omega, \mathcal{F})$ be a measurable space, and μ, μ_1, μ_2, ν are σ -finite measures on it. Define the set function: $\mu_3 \equiv \mu_1 + \mu_2$ (i.e., for all $A \in \mathcal{F}$, $\mu_3(A) = \mu_1(A) + \mu_2(A)$). Then, prove the following:

(a) μ_3 is a measure.

(b) If $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\mu_3 \ll \nu$.

(c) Express the Radon-Nikodym derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$ (proof required).

(d) If $\mu \ll \nu$ and $\mu \perp \nu$ holds, show that μ is the zero measure.

(e) If $\mu_1 = \text{Lognormal}(5, 6)$, $\mu_2 = \text{Geometric}(0.3)$ and $\nu = \text{Poisson}(8)$ on $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then find the Lebesgue decomposition of μ_3 w.r.t. ν , and find the Radon Nikodym derivative $\frac{d\mu_{3a}}{d\nu}$, where μ_{3a} is the absolutely continuous component of μ_3 in the decomposition.

(6+8+8+8+10=40 points)

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

- (1) (a) Let $(\Omega, \mathcal{F}, P) = ([0, 1], \mathcal{B}([0, 1]), m)$, where m denotes the Lebesgue measure on $[0, 1]$. Define the sequence of random variables X_n such that $X_1 \equiv 0$, $X_2 \equiv 13$ and $X_n(\omega) = n\mathbf{I}_{[\frac{1}{2}-\frac{1}{n}, \frac{1}{2}+\frac{1}{n}]}(\omega)$, $\omega \in \Omega$, $n \geq 3$. Let $X \equiv 0$.
- (i) Prove that $X_n \rightarrow X$ a.e. (P) as $n \rightarrow \infty$.
 - (ii) Prove: $\{X_n\}$ **not** uniformly integrable.
 - (iii) Prove: X_n does **not** converge to X in L^1 as $n \rightarrow \infty$.
- (b) In general, under what conditions on the functions $\{X_n\}$ would
 “ $X_n \rightarrow X$ a.e (P)” imply “ $X_n \rightarrow X$ in L^p ” for $p \geq 1$?
 ((3×8)+8=32 points)
- (2) Let $(\Omega, \mathcal{F})$ be a measurable space, and μ, μ_1, μ_2, ν are σ -finite measures on it. Define the set function: $\mu_3 \equiv 2\mu_1 + 3\mu_2$ (i.e., for all $A \in \mathcal{F}$, $\mu_3(A) = 2\mu_1(A) + 3\mu_2(A)$). Then, prove the following:
- (a) μ_3 is a measure.
 - (b) If $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\mu_3 \ll \nu$.
 - (c) Express the Radon-Nikodym derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$ (proof required).
 - (d) Let $\mu = \mu_3$ defined as defined above where $\mu_1 = \text{Normal}(2, 3)$, $\mu_2 = \text{Poisson}(6)$. Find the Lebesgue decomposition of μ with respect to $\nu = m =$ the Lebesgue measure on $\mathbb{R}$, and find the Radon Nikodym derivative $\frac{d\mu_a}{d\nu}$, where μ_a is the absolutely continuous component of μ in the decomposition.
 (4×10=40 points)
- (3) Suppose X is a random variable defined on some probability space $(\Omega, \mathcal{F}, P)$.
- (a) If $M(t) = E(e^{tX})$, $t \in \mathbb{R}$ is the moment generating function of X , then show

$$P(X > a) \leq \inf_{t>0} e^{-ta} M(t), \quad a > 0.$$
 - (b) If $X > 1$ a.e (P), then prove $E(\log X) \leq \log(EX)$.
 (8+10 =18 points)
- (4) (a) **State** precisely the *Dominated convergence Theorem (DCT)* (Not the EDCT!).
- (b) **Prove** that the conclusion of DCT remains true even if “ $f_n \rightarrow f$ a.e(μ)” is replaced by “ $f_n \rightarrow f$ in measure” in the statement.
 (6 + 9 = 15 points)

2016

STAT 641 (Ghosh)

Midterm Exam 2

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Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

- (1) (a) Let $(\Omega, \mathcal{F}, P) = ([0, 1], \mathcal{B}([0, 1]), m)$, where m denotes the Lebesgue measure on $[0, 1]$. Define the sequence of random variables X_n such that $X_1 \equiv 0$, $X_2 \equiv 13$ and $X_n(\omega) = n \mathbf{I}_{\{[\frac{1}{n}, \frac{2}{n}] \cup [1 - \frac{1}{n}, 1]\}}(\omega)$, $\omega \in \Omega$, $n \geq 3$. Let $X \equiv 0$.

- (i) Prove that $X_n \rightarrow X$ a.e. (P) as $n \rightarrow \infty$.
- (ii) Prove: $\{X_n\}$ **not** uniformly integrable.
- (iii) Prove: X_n does **not** converge to X in L^1 as $n \rightarrow \infty$.

- (b) In general, under what conditions on the functions $\{X_n\}$ would
 “ $X_n \rightarrow X$ a.e (P)” imply “ $X_n \rightarrow X$ in L^p ” for $p \geq 1$?

((3×10)+10=40 points)

- (2) Let $(\Omega, \mathcal{F})$ be a measurable space, and μ, μ_1, μ_2, ν are σ -finite measures on it. Define the set function: $\mu_3 \equiv \underline{3\mu_1 + 2\mu_2}$ (i.e., for all $A \in \mathcal{F}$, $\mu_3(A) = 3\mu_1(A) + 2\mu_2(A)$). Then, prove the following:

- (a) μ_3 is a measure.

- (b) If $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\mu_3 \ll \nu$. Also, express the Radon-Nikodym derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$ (proof required).

- (c) Let $\mu = \mu_3$ be defined as above where $\mu_1 = \text{Normal}(2, 3)$, $\mu_2 = \text{Poisson}(6)$. Find the Lebesgue decomposition of μ with respect to $\nu = m =$ the Lebesgue measure on $\mathbb{R}$, and find the Radon Nikodym derivative $\frac{d\mu_a}{d\nu}$, where μ_a is the absolutely continuous component of μ in the decomposition.

(10+(10+10)+15=45 points)

- (3) Suppose X is a random variable defined on some probability space $(\Omega, \mathcal{F}, P)$.

- (a) If $X > 1$ a.e (P), then prove $E(\log X) \leq \log(EX)$.

- (b) Let Z be a *nonnegative* random variable with $0 < \text{Var}(Z) < \infty$ Prove:

$$P(Z > 0) \geq \frac{[E(Z)]^2}{E(Z^2)}.$$

[Use Cauchy-Schwarz Inequality]

(10+10 = 20points)

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

- (1) (a) Let $(\Omega, \mathcal{F}, P) = ([0, 1], \mathcal{B}([0, 1]), m)$, where m denotes the Lebesgue measure on $[0, 1]$. Define the sequence of random variables X_n such that $X_1 \equiv 0$, $X_2 \equiv -5$ and $X_n(\omega) = n^2 \mathbf{I}_{[1-\frac{1}{n^2}, 1]}(\omega)$, $\omega \in \Omega$, $n \geq 3$. Let $X \equiv 0$.
- (i) Prove that $X_n \rightarrow X$ a.e. (P) as $n \rightarrow \infty$.
 - (ii) Prove: $\{X_n\}$ **not** uniformly integrable.
 - (iii) Prove: X_n does **not** converge to X in L^1 as $n \rightarrow \infty$.
- (b) In general, under what conditions on the functions $\{X_n\}$ would
 “ $X_n \rightarrow X$ a.e (P)” imply “ $X_n \rightarrow X$ in L^p ” for $p \geq 1$?

((3×10)+10=40 points)

- (2) Let $(\Omega, \mathcal{F})$ be a measurable space, and μ_1, μ_2, ν are σ -finite measures on it. Define the set function: $\mu_3 \equiv 3\mu_1 + 2\mu_2$ (i.e., for all $A \in \mathcal{F}$, $\mu_3(A) = 3\mu_1(A) + 2\mu_2(A)$). Then, prove the following:

- (a) μ_3 is a measure.
- (b) If $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\mu_3 \ll \nu$. Also, express the Radon-Nikodym derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$ (proof required).

Let $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $\mu = (0.3)\mu_1 + (0.7)\mu_2$, where $\mu_1 = \text{Normal}(2, 3)$ and $\mu_2 = \text{Poisson}(6)$. Let $\nu = m =$ the Lebesgue measure.

- (c) Find the Lebesgue decomposition of μ with respect to ν , and find the Radon Nikodym derivative $\frac{d\mu_a}{d\nu}$, where μ_a is the absolutely continuous component of μ in the decomposition.

- (d) Let X be a random variable with distribution μ (i.e. $P_X = \mu$). Find $E(X^2)$.
 (Hint: a $\text{Normal}(2, 3)$ random variable has mean is 2, variance is 3 and a $\text{Poisson}(6)$ random variable has mean 6 and variance 6).

(10+(10+10)+10+10=50 points)

- (3) Suppose X is a random variable defined on some probability space $(\Omega, \mathcal{F}, P)$. Then show: $X \in L^{p_2}(\Omega, \mathcal{F}, P)$ implies $X \in L^{p_1}(\Omega, \mathcal{F}, P)$, where $0 < p_1 < p_2 < \infty$.
 (Recall: $X \in L^p(\Omega, \mathcal{F}, P)$ means $E[|X|^p] < \infty$)

(15 points)

Write legibly showing all the steps VERY clearly. (Total points = 100).

- (1) Let $(\Omega, \mathcal{F}, P) = ((0, 1), \mathcal{B}((0, 1)), m)$, where m denotes the Lebesgue measure. Define the random variables $X, \{X_n\}$: For each $\omega \in \Omega$, $X(\omega) = 1$ and

$$X_1(\omega) = 0, X_n(\omega) = 1 + (n-1)\mathbf{I}_{(0, \frac{1}{n}]}(\omega), n \geq 2.$$

- (a) Does $X_n \rightarrow X$ a.s. (P) as $n \rightarrow \infty$?
- (b) Does $X_n \rightarrow X$ in probability as $n \rightarrow \infty$?
- (c) Does $X_n \rightarrow X$ in L^p as $n \rightarrow \infty$? Argue for different values of $p \in (0, \infty)$.
- (d) Is $\{X_n\}$ a uniformly integrable sequence of random variables?

(10+10+10+10=40 points)

- (2) Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $g, f, f_n \in L^1(\Omega, \mathcal{F}, \mu) \equiv L^1$ for $n \geq 1$.

- (a) If $f_n \leq f$ then show that $\limsup_{n \rightarrow \infty} \int f_n d\mu \leq \int (\limsup_{n \rightarrow \infty} f_n) d\mu$.
- (b) Assuming $f > 0$ a.e. (μ) and $\frac{1}{f} \in L^1$, show $\left(\int f d\mu\right) \left(\int \frac{1}{f} d\mu\right) \geq 1$.
- (c) If $\int_A f d\mu = \int_A g d\mu$ for all $A \in \mathcal{F}$, then show that $f = g$ a.e. (μ) .

(10+10+10=30 points)

- (3) Let $(\Omega, \mathcal{F})$ be a measurable space and μ, μ_1, μ_2, ν are some σ -finite measures on it.

- (a) If $\mu \ll \nu$ and $\mu \perp \nu$, show: μ is the zero measure, i.e. $\mu(A) = 0$, for all $A \in \mathcal{F}$.

Let $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $\mu_3 \equiv \mu_1 + 3\mu_2$, where $\mu_1 = \text{Normal}(-1, 1)$, $\mu_2 = \text{Bernoulli}(\frac{1}{5})$ and $\nu = m$ is the Lebesgue measure.

- (b) Find the Lebesgue decomposition of μ_3 w.r.t. ν , and find the Radon Nikodym derivative $\frac{d\mu_{3a}}{d\nu}$, where μ_{3a} is the absolutely continuous component of μ_3 in the decomposition.

- (c) If $g(x) = x$, for $x \in \mathbb{R}$, and μ_{3a} is as in (d), find $\int g d\mu_{3a}$.

Give the final answer in numerical form. You can state and use any calculus/basic statistics facts without proofs (but explain the *measure-theoretic* steps.).

(10+10+10=30 points)

1. Let m be the Lebesgue measure on $[0, 1]$ and $I_A(x)$ denote the indicator function of A . Define a sequence of random variables on $(\Omega = [0, 1], \mathcal{F} = \mathcal{B}[0, 1], P = m)$: For $n \geq 1$, $X_n(\omega) = 1 + nI_{[0, \frac{1}{n}]}(\omega)$ and $X(\omega) = 1$ for $\omega \in \Omega$.

(10+10+10+15 = 45 points)

1. (i) Show $X_n \rightarrow X$ almost surely (P).

1. (ii) Does $X_n \rightarrow X$ in probability ?

1. (iii) Is the class $\{X_n\}$ uniformly integrable (U. I)?

1. (iv) Show $X_n \rightarrow X$ in $L^p \equiv L^p(\Omega, \mathcal{F}, P)$ for $0 < p < 1$, but this convergence does not hold for $1 \leq p < \infty$.

- (2) Let X, Y, Z be random variables on some probability space and $1 < p, q < \infty$.
(10+15 = 25 points)
2. (a) Let $Y \in L^p, Z \in L^q$ where $\frac{1}{r} = \frac{1}{p} + \frac{1}{q}$, then show $YZ \in L^r$.

2. (b) Show that $E\left(e^{X^2}\right) \geq e^{E(X^2)}$

(10+15 = 25 points)

- (3) Let $(\Omega, \mathcal{F})$ be a measurable space, and μ, μ_1, μ_2, ν are σ -finite measures on it. Define the set function: $\mu_3 \equiv 3\mu_1 + 2\mu_2$ (i.e., $\mu_3(A) = 3\mu_1(A) + 2\mu_2(A)$ for all $A \in \mathcal{F}$). Show:
(10+(5+5)+10=30 points)

3. (a) μ_3 is a measure.

3. (b) If $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\mu_3 \ll \nu$. Also, express the Radon-Nikodym derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$ (proof required).

3. (c) Let $\mu = \mu_3$ be defined as above where $\mu_1 = \text{Binomial}(10, 0.5)$, $\mu_2 = \text{Normal}(1, 1)$. Find the Lebesgue decomposition of μ with respect to $\nu = m =$ the Lebesgue measure on $\mathbb{R}$, and find the Radon Nikodym derivative $\frac{d\mu_a}{d\nu}$, where μ_a is the absolutely continuous component of μ in the decomposition.

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^1 p^i (1-p)^{1-i} I_A(i)$
Binomial (n, p), $0 < p < 1, n \in \mathbb{N}$	$\sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Uniform (a, b), $-\infty < a < b < \infty$	$m(A \cap [a, b]) / (b - a)$
Exponential (β), $\beta \in (0, \infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$
Gamma (α, β), $\alpha, \beta \in (0, \infty)$	$\int_A \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx, a \in (0, \infty)$
Beta (α, β), $\alpha, \beta \in (0, \infty)$	$\frac{1}{B(\alpha, \beta)} \int_A x^{\alpha-1} (1-x)^{\beta-1} I_{(0, 1)}(x) m(dx),$ where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b), a, b \in (0, \infty)$
Cauchy (γ, σ) $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\pi\sigma} \frac{\sigma^2}{\sigma^2 + (x-\gamma)^2} m(dx)$
Normal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi\sigma}} \exp(-(x-\gamma)^2 / 2\sigma^2) m(dx)$
Lognormal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi\sigma}} \frac{e^{-(\log x - \gamma)^2 / 2\sigma^2}}{x} I_{(0, \infty)}(x) m(dx)$

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